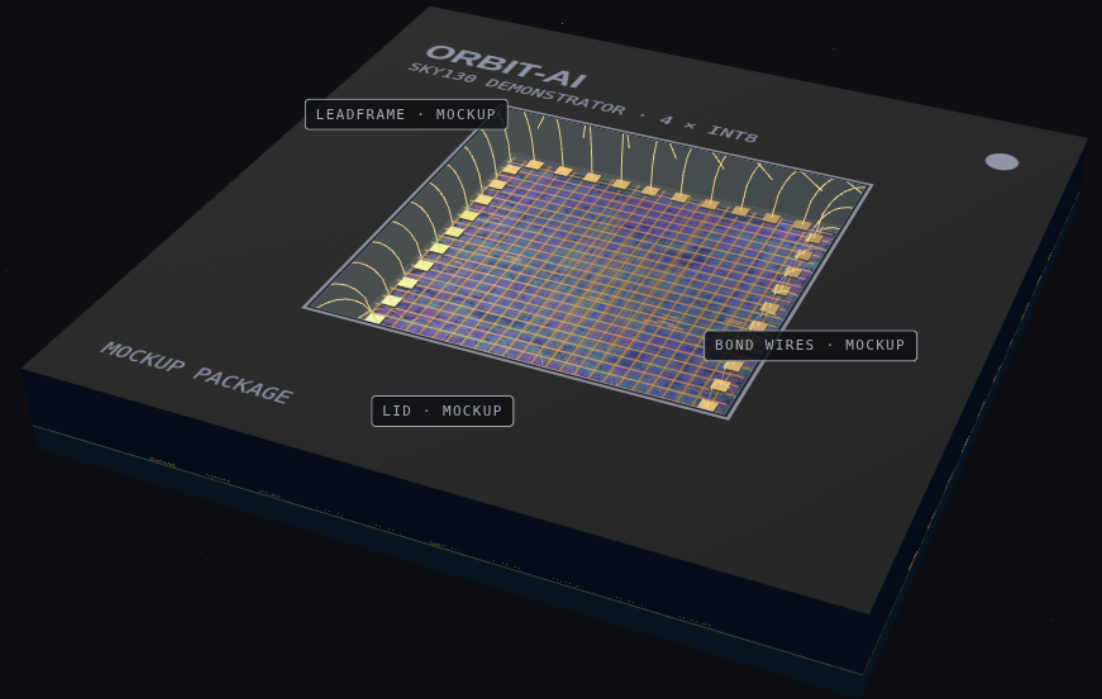


VANTAGE · SPACE COMPUTE

ORBIT-AI

An AI chip built for orbit.

It catches its own radiation errors before they become wrong answers.



Space breaks normal chips.

Satellites need AI on board, but orbit is hostile to the silicon that runs it.

01

Radiation flips bits

A single particle can silently change a result.

02

No repair visits

Whatever launches has to fix itself.

03

Heat only radiates

No air to cool. Hot spots must be managed.

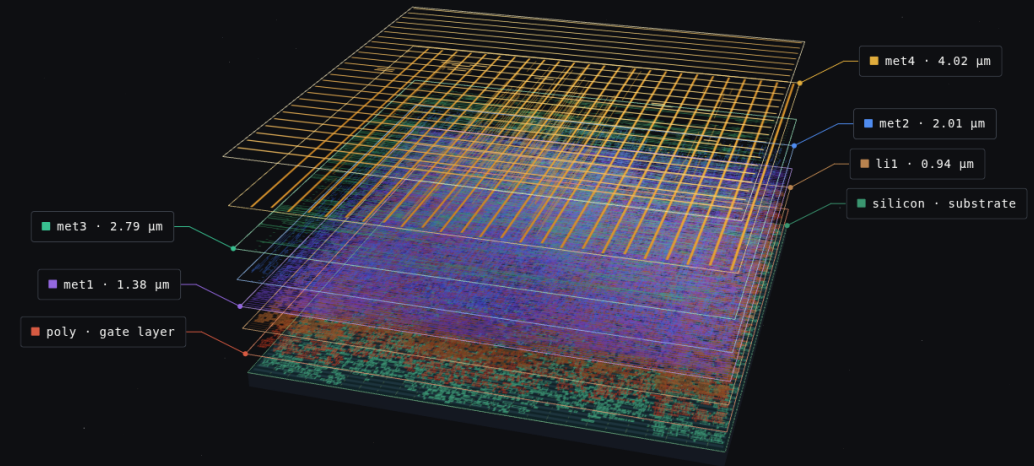
04

Power is scarce

Every watt comes from the solar array.

Built, laid out, and tested.

A 4-lane INT8 AI demonstrator taken from RTL to a sign-off-clean layout with open-source tools on the SkyWater 130 nm process.



346 μm

die edge

139 MHz

clock, timing met

1.1 GOPS

INT8 throughput

4

AI lanes, fully protected

Every result is checked twice.

01

Twin copies

Each lane runs on two copies of the hardware.

02

Compare

The copies are compared every cycle.

03

Fault-stop

A mismatch stops the output. No wrong answer leaves.

04

Triple vote

Thermal control runs in triplicate and votes 2 of 3.

936 bit flips. Zero wrong answers.

Flips injected into the gate-level netlist over 30,000 cycles. Every one was caught.

936 / 936

flips stopped or repaired

702

caught by fault-stop

234

repaired by voting

0

wrong results out

One particle can't hit both copies.

The twin copies sit in fenced regions on opposite sides of the die, so a single strike cannot corrupt both.

$\geq 251 \mu\text{m}$

between twin copies

$242 \mu\text{m}$

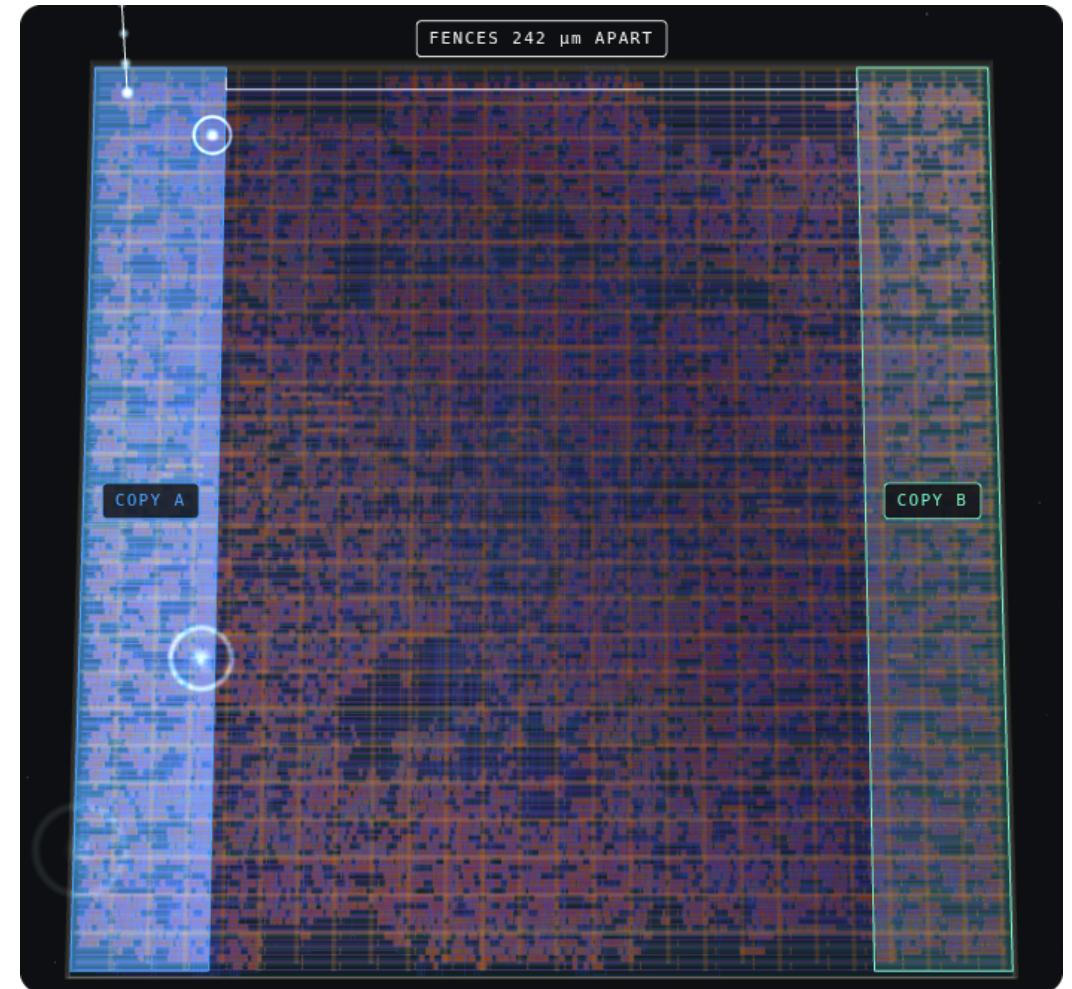
gap between fences

0 / 262

same-bit pairs touching

13 $\rightarrow$ 0

vs. unfenced layout



Clean enough to tape out.

0

DRC violations

Clean

LVS: layout matches netlist

+0.031 ns

timing slack at 7.2 ns

Only ORBIT-AI checks every box.

	ORBIT-AI	RAD750	GR740	XQR Versal	PIC64-HPSC	Jetson Orin NX
AI inference engine (INT8)	✓	—	—	✓	✓	✓
Hardware fault tolerance beyond memory ECC	✓	✓	✓	✓	✓	—
Separated copies, measured in layout	✓	—	—	—	—	—
Fault injection published bit by bit	✓	—	—	—	—	—
No wrong result on a detected upset	✓	—	—	—	—	—
Open design	✓	—	—	—	—	—
	6/6	1/6	1/6	2/6	2/6	1/6

From demonstrator to full-size chip.

01

Built

4-lane demonstrator.
Layout clean, faults
proven in simulation.

02

Full-size design

16 tiles, 6.55 to 26.21
INT8 TOPS. Designed
on paper.

03

Radiation test

Heavy-ion and proton
beam testing of the
protection scheme.

04

Tapeout

First silicon, then flight
demonstration.

Explore the real layout in 3D.

Every block, fence, and copy is labeled from the routed design data.

3D die explorer

`vittesh12345.github.io/chip/
orbit-ai/explorer/`

Interactive story

`vittesh12345.github.io/chip/
orbit-ai/`

Open source

`github.com/vittesh12345/chip`

VANTAGE

AI that stays right in orbit.

Let's talk.

ORBIT-AI